Synchronization and Processes

Dr. Namita Panda
Associate Professor
School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar

Clock Synchronization Algorithms

- Clock Synchronization
- Clock Synchronization Algorithms
- Mutual Exclusion Algorithms
- Election Algorithms
- Atomic Transactions & Modeling
- Atomic Transaction Implementation
- Concurrency Control Algorithms in Atomic Transaction

Basic Concepts:

Goal:

1.If one machine has a WWV(UTC provider used by National Institute of Standard Time(NIST) receiver ,then to keep all other machines synchronized .

2.If no machine have WWV receiver, then each machine keeps track of its own time as well as keep all other machines together w.r.t the average time stamp.

Basic Concepts cont..

Assumtions:

- Each machine hava a timer and its interrupt is *H* times a second.
- When the timer goes off,the interrupt handler adds 1 to a s/w clock(value #C),which keeps track of the number of interrupts.
- When UTC(Universal Coordinated Time) time is *t*, the value of the clock on machine *p is* $C_p(t)$.
- In Perfect world

$$C_p(t)=t \text{ for all } p \text{ and } t$$

OR

$$dC/dt=1$$

Basic Concepts cont..

- Practically real timers do not interrupts exactly *H* times a second.
- If *H*=60 then no of ticks per hour 216,000
- As modrn timer chips have error about 10^{-5},so number of ticks are 215,998 to 216,002 per hour.
- By considering the constant p then

$$1-p <= dC/dt <=1+p$$

(p is known as **maximum drift rate)**

Basic Concepts cont..

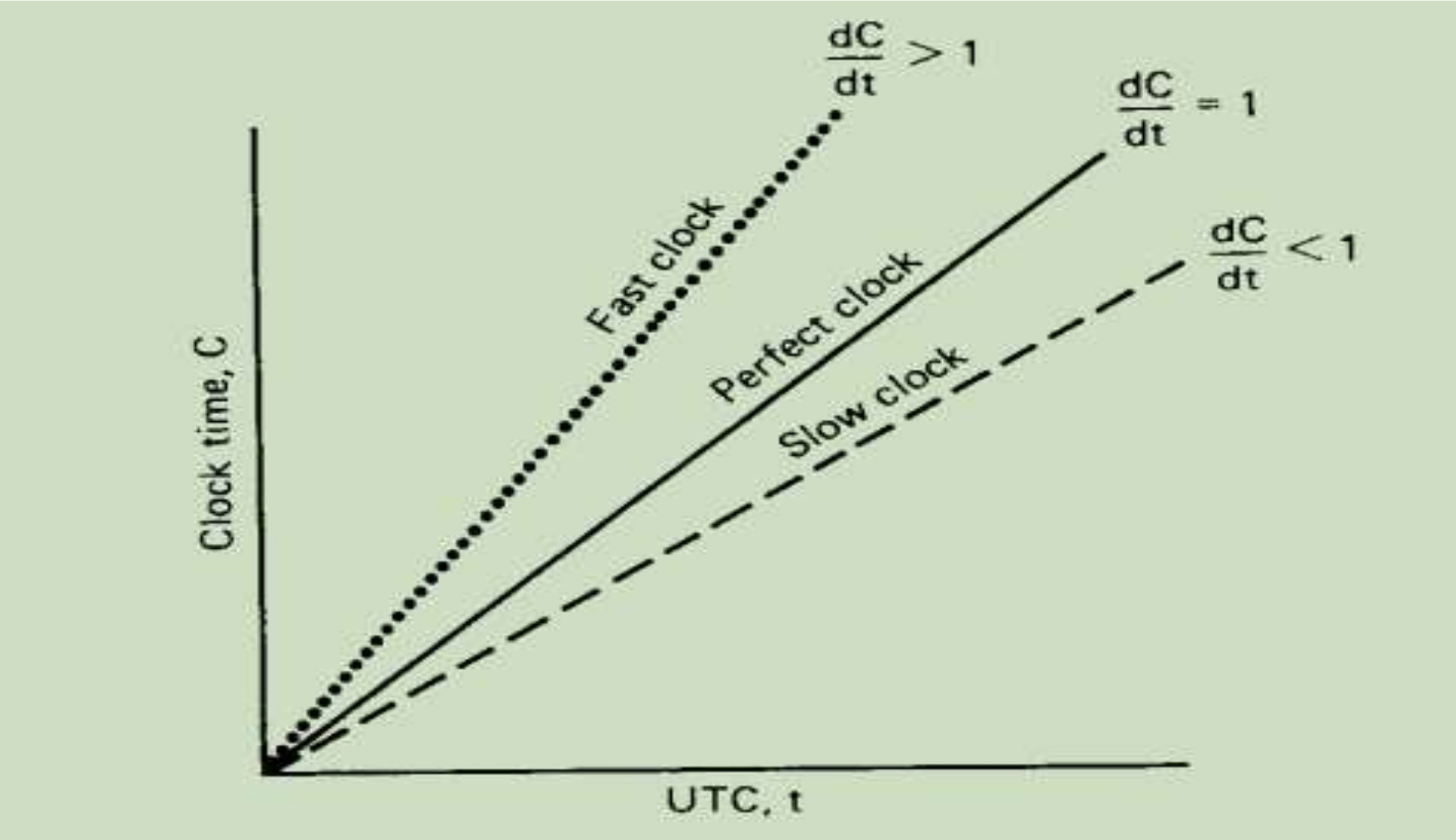


fig: Not all clocks tick precisely at the correct rate

Cristian's Algorithm

- One machine uses WWV receiver considered as **Time server.**
- Goal is to have all other machines stay synchronised.
- Periodically , each machine sends a request message to the time server asking it for the current time.
- Time server responds as fast as it can with a message containing its current time, C_{UTC} shown in following Fig.
- Let **d** is the estimate of delay from the time server to the sender which is also known as **message propagation time.**
- So sender should set its clock to $\mathbf{C_{UTC}+d}$

Cristian's Algorithm cont..

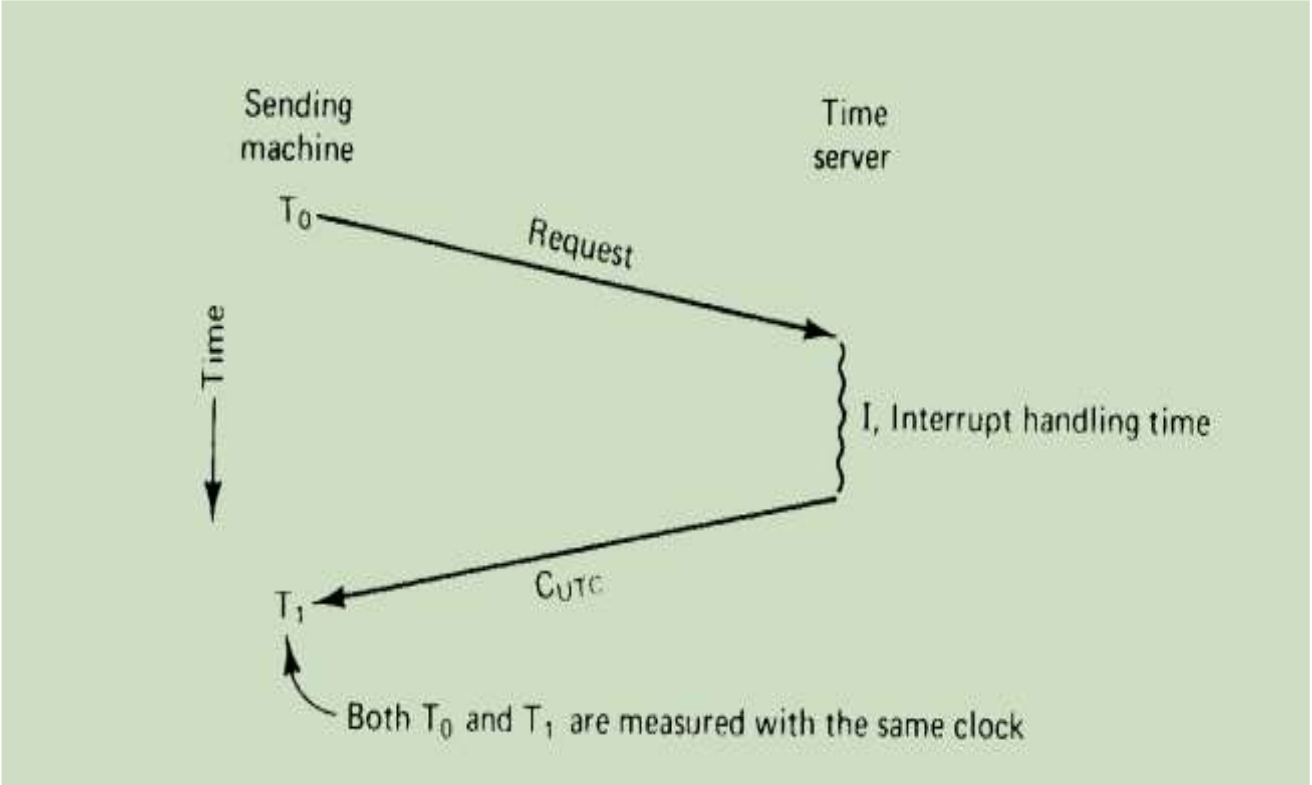


Fig:Getting the current time from the time server

Cristian's Algorithm cont..

- If nothing is known about **I**(interrupt handling time),

  esimated $\mathbf{d=(T_1-T_0)/2}$

  where starting time T_0 and ending timeT_1
- If **I** is known ,then estimated $\mathbf{d=(T_1-T_0-I)/2}$

OR

- **d** may be estimated using **average or minimum** of multiple requests.

Cristian's Algorithm cont..

PROBLEMS OF Cristian's Algorithm:

1.(Major problem)

Time must never run backward.

If the sender's clock is fast, C_{utc} will be smaller than the sender's current value of C which is serious problem.

SOLUTION:

Such change must be introduced gradually,instead of jumping it forward all at once.

EXAMPLE:Suppose the timer is set to generate 100 interrupts per second ,each interrupt would add 10msec to the time and when slowing down,the interrupt routine adds 9msec each time,until the correction has been made.

Cristian's Algorithm cont..

2. (Minor problem)

It takes a nonzero amount of time for the time server's reply to get back to the sender.

SOLUTION:

Sender must record accurately the T_0 and T_1 which are measured using the same clock, so the interval will be relatively accurate.

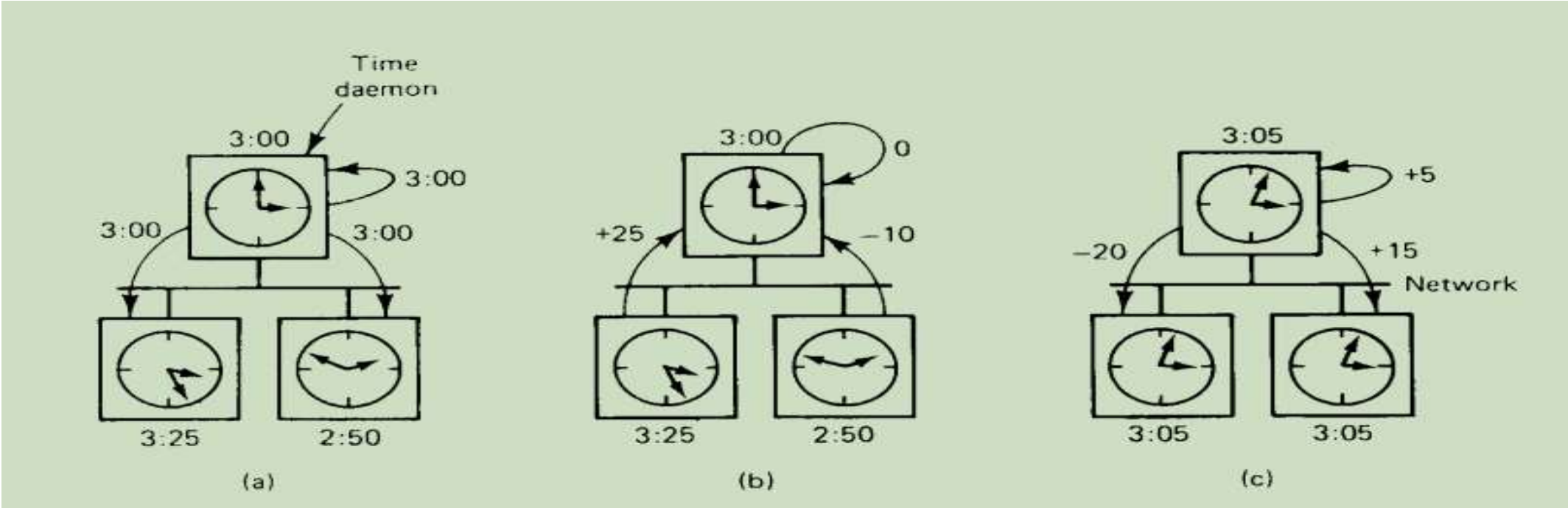
Berkeley Algorithm

- In Cristian's algorithm time server is passive.
- In Berkeley Algorithm time server is active known as Time Daemon.
- In this case no machine has a WWV receiver.
- The time server periodically gathers time data from all other machines by polling technique and determine the average time. Then it communicates the average time (in fact the difference) to all machine to advance their clocks to the new time or slow their clocks down until some reduction has been achived.

Berkeley Algorithm cont..

- The time daemon's time must be set mannually by the operator periodically.

Fig(a)the time daemon asks all other machines for their clock value.
(b)The machines answer
(c)The Time daemon tells everyone how to adjust their clock



Averaging Algorithms:

- This algorithm is known as decentralised algorithm.
- Devide the time into fixed length resynchronization interval
- The *i*th interval starts at $T_0 + iR$ and runs until $T_0+(i+1)R$
- T_0 is agreed upen moment on past and R is system parameter.
- At the beginning of each interval, every machine **broadcasts the current time** according to its clock as different clock has different speed.
- Then it **starts a local timer to collect all other broadcasts** during some interval S.
- The algorithm runs to **compute a new time from them by averaging the values from other machines**.

Averaging Algorithms cont...

Variations:

1.**Discard the *m* highest and *m* lowest values** and average the rest,as the extreme values consider as faulty clocks.

2. Collect each message **including its propagation time** from the sources.

Use of synchronized clocks

1. At most once message delivery:

- Traditional approch:
- Each messsage has an unique number and server stores all the numbers of message and can detect new messages from retransmission.
- Problem:
- If server crashes and reboots it loses its table of message numbers.
- Modified approch(With time):
- Each message carries a connection ID and timestamp.
- for each connection server records in table the most recent timestamp it has seen,if message for a connection is lower than the timestamp stored for that connection,is rejected.

Use of synchronized clocks cont..

2.**Clock based cache consistency**

- It is used in distributed file system.
- Traditional approch:
- Each client maintains local cache which is partitioned as read and write file.
- Problem :
- If a client has a file cached for reading,before another client can get a copy for writting,the the server has to first ask the reading client to invalidate its copy.
- Solution for the extra overhead can be eliminated by synchrinized clock

Use of synchronized clocks cont..

- Modified approch(With synchronized clock):
- When a client wants a file,it is given a **lease** on it which specifies how long the copy is valid.
- If the lease expires,the client can ask for it to be renewed.
- If the lease expires and the file is needed again shortly then the client can ask the server if the copy it has(identified by timestamp) is still the current one then a new lease is generated,but the file need not be retransmitted.

Q&A

QUESTIONS:

Q1.Consider the behavior of two machine in a distributed system .Both have clocks that areSupposed to tick 1000 time per milisecond. One of them actually does,but the other ticks only 990times per milisecond.If UTC updates come in once a minute,what is the maximum clock skew that will occur?

Q2.Discuss the problems and solutions of **Cristian's Algorithm.**

Q3.How diffrent clocks are synchronised using Berkeley Algorithm?

Q4.In the approch to cache consistency using leases,is it really essential that the clocks are synchronized?

Thank You!